$\boxed{\text{One proportion}: \hat{p} \text{ vs } p}$ Hypothese testing

* A 2013 Research Poll found 60% of 1983 randomly sampled American adults believe in evolution. Does this provide convincing evidence that majority of Americans believe in evolution?

$$Z = \frac{\hat{p} - p}{SE}, \quad SE = \sqrt{\frac{p(1-p)}{n}}$$

$H_0: p = 0.5$

$H_A: p > 0.5$

$\hat{p} = 0.6$

$n = 1983$

* condition:

① independence: 1983 < 10% of Americans & randomly sampled ✓

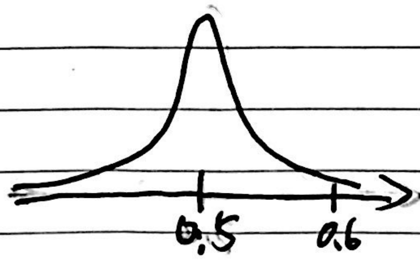
② sample size / skew

$np = 1983 \times 0.5 = 991.5 > 10$ ✓

→ nearly normal sampling distribution!

$$\hat{p} \sim N\left(\text{mean} = 0.5,\ SE = \sqrt{\frac{0.5 \times 0.5}{1983}} \approx 0.0112\right)$$

note here we use $p = 0.5$



$$Z = \frac{0.6 - 0.5}{0.0112} \approx 8.92$$

$$p\text{-value} = P(Z > 8.92) \sim 0$$

→ reject H_0

p-value = P(observed or more extreme outcome | null hypothes true)

⇒ There is almost 0% chance of obtaining a random sample of 1983 Americans where 60% or more believe in evolution, if in fact 50% of Americans believe in evolution.

Two proportion : $\hat{p}$ vs p Hypothese testing

* A surveyUSA poll asked respondents whether any of their children have ever been the victim of bullying. Also recorded on this survey was the gender of the respondent. Below is the distribution of respondents by gender:

	Male	female
Yes	34	61
No	52	61
not sure	4	0
total	90	122
$\hat{p}$	0.38	0.50
	= 34/90	= 61/122

$H_0: P_{male} = P_{female}$, $P_{male} - P_{female} = 0$

$H_A: P_{male} \neq P_{female}$, $P_{male} - P_{female} \neq 0$

$$\hat{P}_{pool} = \frac{34+61}{90+122} = 0.45$$

Condition:

① Independence: within group: random sample, <10% ✓
between groups

② sample size/skew: $\hat{P}_{pool}$

male: $90 \times 0.45 = 40.5$ $1 - \hat{P}_{pool}$
$90 \times 0.55 = 49.5 > 10$

female: $122 \times 0.45 = 54.9 > 10$ ✓

⇒ success-failure conditions for male & female are met.

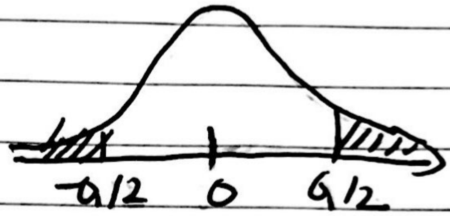
⇒ difference between two sample proportions is nearly normal.

$$(\hat{p}_{male} - \hat{p}_{female}) \sim N\left(mean = 0,\ SE = \sqrt{\frac{0.45\times0.55}{90} + \frac{0.45\times0.55}{122}} \approx 0.0691\right)$$

point estimate $= \hat{p}_{male} - \hat{p}_{female} = 0.38 - 0.5 = -0.12$

null value $= 0$

$H_0: \hat{p}_{male} - \hat{p}_{female} = 0$



$$Z = \frac{(-0.12) - 0}{0.0691} \approx -1.74$$

p-value $= P(|Z| > 1.74) \approx 0.08$

$\Rightarrow$ fail to reject H_0

$\Rightarrow$ there is <u>NO</u> difference in males and females with respect to likelihood of reporting their kids being bullied.